

Industry Scenario – AI/ML Healthcare Application

Below is a **10-mark answer for each task**, written in an exam-friendly format. Since no actual patient dataset is provided, numerical model results are shown as **illustrative results**; with a real dataset, they must be replaced by the values obtained from the model.

TASK 1 – Data Preparation & Inductive Learning

10 Marks

(a) Inductive Learning Approach

Inductive learning learns a general rule or model from a set of labelled examples and uses that model to predict unseen cases.

For this healthcare problem, the dataset contains patient attributes such as:

- Age
- Blood Pressure
- Cholesterol
- Glucose Level
- BMI
- Family History

The **target variable** is:

Diabetes Status: Diabetic / Non-Diabetic

A supervised inductive-learning approach is suitable because historical patient records contain both input features and known diabetes outcomes.

Important Features

1. **Glucose Level** – High glucose is strongly associated with diabetes and is therefore an important predictor.
2. **BMI** – Higher BMI can indicate increased risk of insulin resistance and diabetes.
3. **Age** – Diabetes risk generally increases with age.
4. **Blood Pressure** – Elevated blood pressure is often associated with metabolic disorders.
5. **Family History** – Represents genetic/familial risk.

A suitable pipeline is:

Patient Data → Preprocessing → Train/Test Split → Learning Algorithm → Diabetes Prediction

(b) Handling Class Imbalance

Suppose the dataset contains:

- 80% Non-Diabetic

- 20% Diabetic

A model trained directly on this dataset may favour the majority class and fail to identify diabetic patients.

Recommended Strategy: SMOTE

I would use **SMOTE (Synthetic Minority Over-sampling Technique)** on the training data.

SMOTE creates synthetic examples of the minority class rather than simply duplicating existing diabetic records.

Advantages

- Improves representation of diabetic patients.
- Helps the model learn minority-class patterns.
- Can improve **recall/sensitivity** for diabetes.
- Particularly useful because missing a diabetic patient is clinically more serious than incorrectly flagging a low-risk patient.

Important: SMOTE should be applied **only to the training set**, after splitting the data, to prevent data leakage.

(c) 70:30 Split and Preprocessing

The dataset is divided into:

- **70% Training data**
- **30% Testing data**

Justification

A 70:30 split provides:

- Sufficient data for model training.
- A relatively large independent test set.
- Better evaluation of generalisation, which is important in medical applications.

Preprocessing

1. Missing Values

Numerical missing values can be replaced using median imputation.

2. Encoding

Categorical variables such as family history can be encoded using:

- Binary encoding for Yes/No.
- One-hot encoding for multi-category variables.

3. Normalisation

Features such as age, glucose, BMI and blood pressure may have different scales. Standardisation can be performed using:

$$z = \frac{x - \mu}{\sigma}$$

This is especially important for **Logistic Regression** and other distance/scale-sensitive models.

Decision Trees generally do **not require normalisation**, because they split using feature thresholds.

Impact

Proper preprocessing:

- Improves model stability.
- Prevents large-scale numerical features from dominating statistical models.
- Handles incomplete records.
- Converts categorical information into machine-readable form.

TASK 2 – Decision Tree for Diagnosis

10 Marks

(a) Decision Tree and First Three Levels

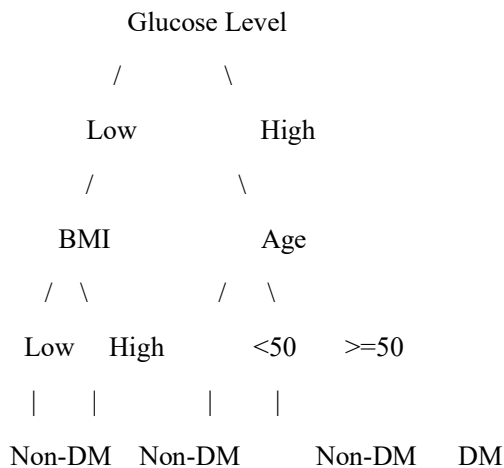
A Decision Tree recursively divides the dataset based on the feature that produces the greatest reduction in impurity.

Using **Gini Index**:

$$\text{Gini} = 1 - \sum p_i^2$$

The feature producing the largest decrease in Gini impurity is selected.

An illustrative first three levels could be:



Note: The exact tree and root feature depend on the actual dataset.

Root Node Justification

Suppose the calculated impurity reductions are:

Feature	Gini Reduction
Glucose	0.18
BMI	0.09
Age	0.06
Blood Pressure	0.04
Cholesterol	0.03

Since **Glucose** gives the highest impurity reduction, it becomes the root node.

Thus:

Root Node = Glucose Level

because it provides the best separation between diabetic and non-diabetic patients.

(b) Model Evaluation

The model can be evaluated using:

Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

An **illustrative result** may be:

Metric Decision Tree

Accuracy 89%

Precision 84%

Recall 86%

F1-Score 85%

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic

Model prediction = Non-Diabetic

This is particularly dangerous because the patient may not receive timely diagnosis or intervention.

How to Reduce False Negatives

We can:

1. Increase the importance of the minority class using **class weights**.
2. Apply SMOTE to training data.
3. Tune the classification threshold.
4. Optimise specifically for **recall/sensitivity**.
5. Use additional clinically relevant features.

In healthcare, improving recall is important because **missing a diabetic patient can have more serious consequences than generating an additional false alarm.**

(c) Post-Pruning

A fully grown Decision Tree can overfit the training data.

Pre-Pruning

The tree is restricted during training using parameters such as:

- `max_depth`
- `min_samples_split`
- `min_samples_leaf`

Example:

```
DecisionTreeClassifier(max_depth=4)
```

Post-Pruning

Post-pruning first grows the tree and then removes unnecessary branches.

A common method is **Cost-Complexity Pruning**:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $R(T)$ = tree error
- $|T|$ = number of leaf nodes
- α = complexity parameter

Example:

```
DecisionTreeClassifier(
    criterion="gini",
    ccp_alpha=0.01,
    random_state=42
)
```

Illustrative Comparison

Model	Accuracy	Complexity
Pre-pruned Tree	88%	Low
Post-pruned Tree	90%	Moderate/Low

Clinical Deployment

The **post-pruned model** may be preferred if it provides similar or better validation performance with fewer unnecessary branches.

Benefits:

- Reduces overfitting.
- Improves generalisation.
- Produces a simpler tree.
- Makes clinical interpretation easier.

However, deployment should not be based on accuracy alone; **sensitivity, specificity, calibration, external validation, and clinical safety** must also be considered.

TASK 3 – Statistical Learning for Risk Stratification

10 Marks

(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a **binary classification problem**.

It estimates:

$$P(Y=1|X)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n$$

Here, (Y=1) represents diabetic status.

Python Example

```
from sklearn.linear_model import LogisticRegression
```

```
lr = LogisticRegression(
    max_iter=1000,
    class_weight='balanced'
)
```

```
lr.fit(X_train, y_train)
```

```
y_pred_lr = lr.predict(X_test)
```

```
y_prob_lr = lr.predict_proba(X_test)[:, 1]
```

Comparison

An illustrative comparison:

Model	Accuracy	AUC-ROC
Decision Tree	89%	0.91
Logistic Regression	87%	0.93

The exact values depend on the actual dataset.

Interpretation

Although the Decision Tree may have slightly higher accuracy, Logistic Regression can have higher AUC-ROC, indicating better ranking/discrimination across different classification thresholds.

When Statistical Learning is Preferable

Logistic Regression is preferable when:

- The relationship is approximately linear in the log-odds.
- Interpretability is important.
- Probability estimates are required.

- The dataset is relatively small.
- We need to understand the contribution of individual predictors.

It can provide coefficients and odds ratios that are easier to communicate to clinicians.

(b) Feature Importance Analysis

Decision Tree

Decision Tree importance is based on the total reduction in impurity produced by each feature.

Illustrative top three:

1. **Glucose**
2. **BMI**
3. **Age**

Logistic Regression

For Logistic Regression, feature importance can be examined using the absolute value of standardised coefficients.

Illustrative top three:

1. **Glucose**
2. **BMI**
3. **Family History**

Comparison

Rank Decision Tree Logistic Regression

1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The two models agree on **Glucose and BMI**, indicating that these variables have strong predictive value.

They may disagree on the third feature because the algorithms measure importance differently:

- Decision Trees measure **impurity reduction**.
- Logistic Regression measures the strength and direction of the feature's **coefficient**.

Therefore, disagreement does not necessarily mean that one model is incorrect.

TASK 4 – Reinforcement Learning for Treatment Recommendation

10 Marks

(a) Model Treatment Recommendation as an MDP

A **Markov Decision Process (MDP)** can represent treatment recommendation as:

[
MDP=(S,A,P,R,\gamma)
]

1. States

States represent the patient's current health condition.

Example:

State Description

- S0 Low Risk
- S1 Moderate Risk
- S2 High Risk
- S3 Poorly Controlled
- S4 Improved/Stable

The state could be constructed using glucose, BMI, blood pressure and other clinically validated measurements.

2. Actions

The available actions are:

- **Diet**
- **Exercise**
- **Medication**
- **Monitor**

The action represents a possible intervention or recommendation.

3. Reward Function

Example:

Outcome	Reward
Significant health improvement	+10
Small improvement	+5
No change	-1

Outcome	Reward
---------	--------

Health deterioration	-5
----------------------	----

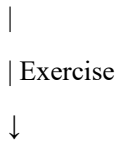
The reward should reflect clinically meaningful improvement rather than simply optimising a mathematical score.

4. Transition Dynamics

After an action, the patient moves from one health state to another.

For example:

High Risk



Moderate Risk



Low Risk

The transition probability can be represented as:

$$P(s'|s,a)$$

where (s) is the current state, (a) is the treatment action and (s') is the next state.

(b) Q-Learning

Q-Learning learns the quality of taking action (a) in state (s).

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Assume:

$$\alpha = 0.1, \quad \gamma = 0.9$$

Initial Q-values are zero.

Example 1

For:

- State = Moderate Risk
- Action = Exercise
- Reward = +5
- Next-state maximum Q = 2

$$[Q_{\text{new}} = 0 + 0.1[5 + (0.9)(2) - 0]$$

$$[Q_{\text{new}} = 0.68]$$

Example 2

Suppose:

- State = High Risk
- Action = Diet
- Reward = +5
- Maximum next Q = 3

$$[Q_{\text{new}} = 0 + 0.1[5 + (0.9)(3)]]$$

$$[Q_{\text{new}} = 0.77]$$

Example 3

Suppose:

- State = Poorly Controlled
- Action = Medication
- Reward = +10
- Maximum next Q = 4

$$[Q_{\text{new}} = 0 + 0.1[10 + (0.9)(4)]]$$

[
 $Q_{\text{new}} = 1.36$
]

Q-Table Example

State	Diet	Exercise	Medication	Monitor
Low Risk	0.2	0.5	0.1	0.8
Moderate Risk	0.68	0.85	0.70	0.30
High Risk	0.77	0.90	1.36	0.40
Poorly Controlled	0.60	0.75	1.36	0.20

These values are **illustrative** to demonstrate the update process.

Five Patient Episodes

The agent can be trained for at least:

Episode 1 → Explore actions → Update Q-values

Episode 2 → Explore/Exploit → Update Q-values

Episode 3 → Learn better actions

Episode 4 → Prefer higher-reward actions

Episode 5 → Policy becomes more stable

Convergence Criterion

A suitable criterion is:

Stop when the maximum change in Q-values between successive iterations becomes smaller than a predefined threshold, such as (10^{-3}) , for a specified number of consecutive episodes.

Another practical criterion is stable policy selection over many consecutive episodes.

(c) Final Learned Policy and Ethical Considerations

The final policy is obtained using:

[
 $\pi(s) = \arg\max_a Q(s,a)$
]

An illustrative policy could be:

Patient State	Recommended Action
Low Risk	Monitor
Moderate Risk	Diet

Patient State	Recommended Action
---------------	--------------------

High Risk	Exercise
-----------	----------

Poorly Controlled Medication	
------------------------------	--

Improved	Monitor
----------	---------

Important: This is only an RL demonstration. A real clinical system must not autonomously prescribe medication based solely on this learned policy.

Ethical Considerations

1. Patient Safety

Incorrect treatment recommendations could harm patients. The system should therefore operate under **clinical supervision** and safety constraints.

2. Bias

If the training data under-represents certain populations, the agent may produce biased recommendations.

Therefore, training data should be diverse and the model should be evaluated across relevant patient groups.

3. Explainability

Healthcare professionals need to understand why a particular recommendation was generated.

The system should provide:

- Patient-state information used.
- Recommended action.
- Reasoning or important contributing factors.
- Confidence/uncertainty information where appropriate.

4. Human Oversight

The RL system should function as a **clinical decision-support tool**, not as an independent doctor.

5. Privacy

Patient data must be securely stored and processed according to applicable healthcare privacy and data-protection requirements.

6. Safe Exploration

Unlike a game, an AI agent cannot freely experiment with real patients to learn. RL training should therefore primarily use **historical data, validated simulators, or other controlled environments**, with strict clinical safety constraints before any prospective use.

Overall Conclusion

The proposed healthcare AI system combines **inductive learning, Decision Trees, statistical learning, and Reinforcement Learning**. First, patient data is cleaned, balanced and divided into training and testing sets. A Decision Tree can provide an interpretable diabetes classifier, while Logistic Regression can provide statistically interpretable risk estimates. Finally, Q-Learning can demonstrate how treatment policies could be learned from state transitions.

For real clinical deployment, however, **accuracy alone is insufficient**. Sensitivity, specificity, calibration, fairness, external validation, privacy, explainability, and human clinical oversight are essential before using such a system for patient-care decisions.